

LEVERAGING HETEROPERCEPTION IN COLLABORATIVE EMBODIED DESIGN: THE CASE OF WHOLE-BODY ENACTING A GEOMETRY DEFINITION

Ratih A. Apsari & Dor Abrahamson

University of California, Berkeley

A group of four 5th-grade participants in a design-based research study that evaluated body-scale geometry activities (n=76) formed a circle in response to a dance task prompt. Drawing on coordination dynamics and ethnomethodological conversation analysis, we examine microgenetically how, without a shared plan, students self-organized via micro-adjustments. The analysis exposes the group's heteroperception (i.e. multiple divergent perspectives) evidenced as conflicting attentional anchors (i.e. perceptual structures for action). Initially implicit, heteroperception became explicit through discussion and resolved as perspectival synergy that promoted geometry discourse. Visualizing heteroperception enables researchers to implicate the emergent coalescence of collaborative enactment.

INTRODUCTION

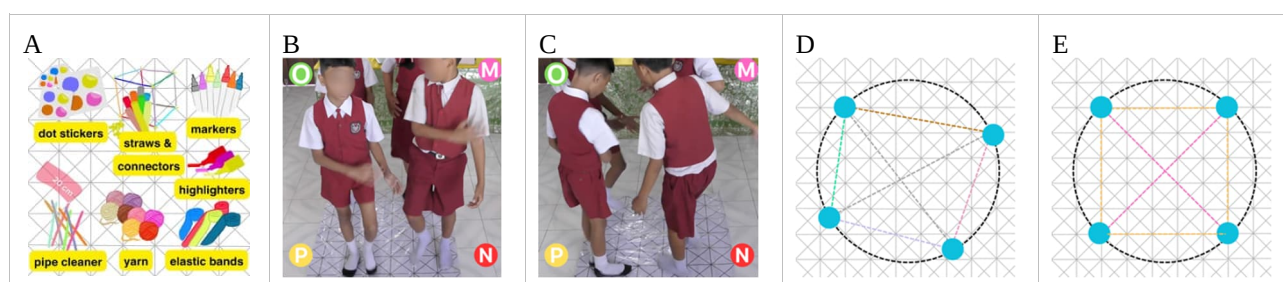
Most students enter mathematics classrooms already familiar with basic geometric entities: they can name, recognize, and draw shapes that count as valid examples. Thus, when teachers request students to *define* well-known shapes, students do not appreciate any epistemic urgency (Herbst et al., 2005). In mainstream mathematics curricula, students' lack of perceived need is often compounded by how definitions are introduced. Geometric definitions are commonly presented as finished products, statements to be memorized or applied, rather than as outcomes of inquiry and problem-solving (Lockhart, 2009). When instruction privileges canonical representations and pre-given definitions, students may come to rely on prototypical appearances rather than on defining relations. This tendency is reflected in persistent difficulties with the hierarchical classification of quadrilaterals (Sinclair et al., 2016) and in recognizing a right angle when it appears in noncanonical orientations (Apsari & Abrahamson, 2024).

In response to these epistemic challenges, researchers have increasingly turned to pedagogical perspectives that foreground how mathematical meaning can emerge from experience rather than being given. For example, Herbst et al. (2005) argue for engaging students in generative activities, in which learners must produce or distinguish figures based on stipulated conditions. A more radical response comes from education scholars who interrogate mainstream pedagogy's implicit epistemological premises by considering alternative philosophical doctrines. Among these, the embodied turn in cognitive science has been quite influential, reshaping how

mathematical thinking is understood, reconceptualizing it as grounded in sensorimotor activity, material engagement, and social interaction. From this perspective, cognition is not confined to internal mental representations but unfolds through coordinated action with others and with the environment (Wilson, 2002). Educational researchers have taken up this view through embodied design (Abrahamson et al., 2020; Alberto et al., 2022; Smith & Walkington, 2020), creating learning environments in which bodily movement is not merely illustrative but constitutive of mathematical reasoning. Studies of collaborative movement tasks demonstrate that whole-body coordination can surface perceptual, spatial, and relational structures for exploration, often revealing mathematical properties that would remain opaque when introduced as a priori concepts couched in small, static two-dimensional images. For instance, design-based researchers have evaluated the learning effects of forming wave ensembles to explore symmetries and transformations (Vogelstein et al., 2019) or manipulating large concrete artifacts to study quotient groups (Lajos et al., 2025).

In a similar spirit, our Geometry Resources in Dance (GRiD) project comprises a suite of whole-body movement activities situated within a dance tradition. Dancers often spontaneously generate imaginary figurative elements to guide their bodily movement and orientation (Clark & Ando, 2014). From an enactive perspective, Hutto and Sánchez-García (2015) conceptualize such perceptual features that guide movement coordination as *attentional anchors* (hence, “AA”). Yet, individuals engaged in enacting an identical choreography may entertain different AA’s (Abrahamson & Shulman, 2019), collectively manifesting *heteroperception*—a socio-cognitive phenomenon in which individuals engaged in joint activity pursue a shared goal while implicitly perceiving and organizing available resources differently (Abrahamson et al., 2025; Benally et al., 2022). In this study, we sought to evaluate whether GRiD participants’ AAs were uniform or, in the case of evident heteroperception, how their perspectival diversity might have served as a generative resource for both constructing and defining a circle.

Figure 1: GRiD activity material resources and the task design



DESIGN

We situated the design within Balinese dance, whose rigorous coordination demands have historically required intergenerational scaffolding of implicit enactive structures; structures that, in turn, are deeply rooted in Balinese embodied, spatial, spiritual, social, engineering, and design practices (Apsari & Abrahamson, 2025). When our study

participants perform a set of assigned choreographic prompts on the GRiD's diagrammatic floor mat (Figure 1A), they *each* spontaneously appropriate the mat as a symbolic artifact enhancing their own epistemic and pragmatic actions—the mat consequently emerges as a means of multimodally coordinating their *joint* action: students' AAs become externalized (Kirsh, 2010) and objectified (Radford, 2021) as a shared frame of reference, enabling participants to reflect on and to “language” geometrically the forms they enact (Apsari & Abrahamson, 2024).

Echoing Chorney (2017), our design aims to position recognizable mathematical objects, such as a circle, as residual cognitive traces of coordinated movement, whereby their formal elements (e.g., center, diameter) come forth through purposeful action. Our 5th-grade Balinese participants were familiar with the circle's shape but had not yet studied its formal elements or definitions. The design leveraged students' familiarity by asking them to form a circle together in groups of four on the GRiD mat, thereby hoping to direct their attention to its constitutive geometric relations.

Beginning in a two-by-two arrangement facing the camera (Figure 1B), students were to perform stylized Balinese walking patterns (*mapal* or *ngegol*, respectively associated with male or female roles) and then turn inward to form a circle (Figure 1C). From a mathematical perspective, four individuals collectively forming a circle may be positioned anywhere along its circumference, configuring an inscribed quadrilateral (Figure 1D). However, the cultural context of traditional dance was to orient students implicitly toward maintaining equidistant spacing along the circumference, resulting in the construction of an inscribed square (Figure 1E).

THEORETICAL FRAMEWORKS

To account for students' mathematical activity as it unfolds through collective, embodied action, we draw on coordination dynamics theory and ethnomethodological conversation analysis (EMCA). From a coordination dynamics perspective, mathematical learning in multi-person action-based inquiry activities is understood as the non-linear self-organization of a functional dynamic system composed of bodies and artifacts multimodally negotiating agencies and intentionalities (Shvarts & Abrahamson, 2023). Conceptual understanding, in this view, is grounded and enacted as goal-directed emergent sensorimotor coordination rather than mediated by abstract mental representations. What learners come to know is inseparable from how they move, orient, and coordinate with others and with material environments.

To conceptualize how such coordination is accomplished in interaction, we adopt the lens of ethnomethodological conversation analysis (EMCA). EMCA treats action—subsuming language, as the primary unit of analysis—attending to how participants organize activity moment by moment through multimodal resources, including talk, gesture, posture, gaze, spatial positioning, and engagement with material artifacts (Mondada, 2019). Crucially, EMCA does not assume a predefined hierarchy among these resources; instead, it investigates how they are dynamically assembled in situated action to establish, negotiate, and repair interlocutors' mutual intelligibility. In

mathematics teaching and learning, mathematical meaning develops as participants render their perceptual experiences public and progressively stabilize shared understandings within designed fields of action.

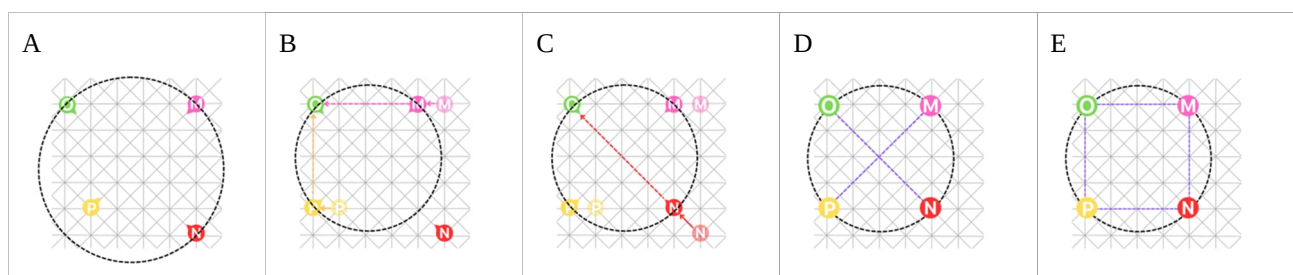
METHOD

This study is part of a design-based research project aimed at developing a culturally situated, embodied mathematical activity. Following multiple rounds of piloting, we conducted and recorded a 60-minute session teaching experiment (Steffe & Thompson, 2000) in Bali with 76 fifth-grade students (ages 9–12). To illustrate the core analytic claims of this paper, we focused on a typical group of four students and examined how they configured a circle on the GRiD floor mat in the absence of an explicit shared plan. As we analyzed the fine-grained organization of participants' interaction from their respective perspectives, differences in their perceptual intentionalities became salient. Capturing how such differences emerge, surface, and are resolved requires close attention to change over short timescales. Accordingly, we employed microgenetic analysis to focus on brief periods of rapid transformation, using dense observations to trace how new forms of coordination arose, fluctuated, and stabilized over the course of activity. To enhance the reliability and interpretive rigor of our analyses, we discussed selected data vignettes among a dedicated group of scholars working on embodied mathematics design.

RESULTS AND DISCUSSION

The focal group—Merry (M), Nath (N), Owen (O), and Philip (P)—walked silently toward one another, continuously micro-adjusting their positions relative to their peers, without an explicit shared plan or designated leader, until a circle formed and stabilized (Figure 2A–C). Although the group was clearly oriented toward a common goal of forming a circle, post-task reflection revealed that their local adjustments were guided by different AAs. Specifically, some students were oriented toward relations between opposite positions (*opposites*; Figure 2D), whereas others attended to adjacent relations (*adjacents*; Figure 2E). These variations in AA were initially implicit, then became explicit during discussion, and ultimately were resolved.

Figure 2: Overhead schematized depiction of students' microadjustments and spontaneous strategies for creating a circle



Note. Panels A–C: Dots with initials represent students' mat positions; small triangles ("noses") show body orientation. Faded dots mark prior positions, solid arrows show movement, dashed arrows foreshadow the content of students' verbal and non-verbal explanations. Panels D–E: Dashed purple lines show students' gazes as they assess spatial orientation toward their peers positioned as either *opposites* (Panel D, X-form) or *adjacents* (Panel E, L-forms from each point's perspective).

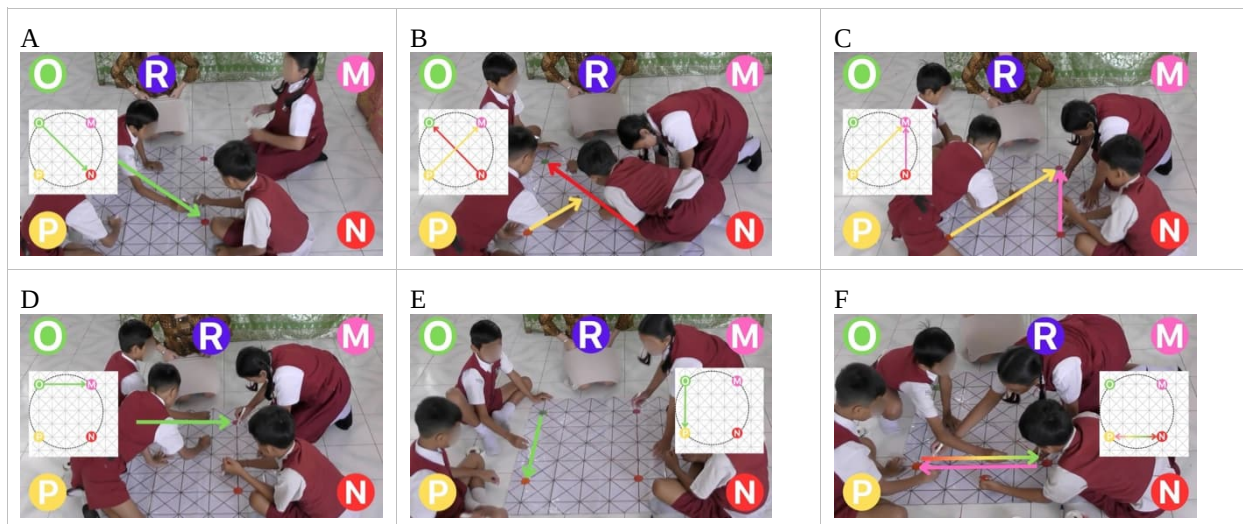
Implicit. Asked to mark their positions on GRiD, students assumed Tapak Sirang Pada, a Balinese V-shaped foot posture, in which heels meet, toes rotated outward (Apsari & Abrahamson, 2024). The vertex of the V indexed the students' location on the mat, which they marked with a sticker. Merry lost her spot while reaching for a sticker, then recalibrated using Owen's position, revealing her use of the *adjacents* strategy (Figure 3A).

Figure 3: Students' implicit and explicit orientational reference points



Explicit. In the follow-up discussion, Nath articulated the *opposites* strategy by gesturing and saying “face-to-face” (Figure 3B). Simultaneously, Philip traced a line toward Owen, reflecting his use of *adjacents* (Figure 3B). Upon hearing Nath, Philip stopped, while Merry rapidly shifted her gaze between Owen and Philip—both signaling a breakdown, realizing they had been entertaining different strategies. Merry and Philip then substituted *adjacents* with *opposites*, illustrating perspectival replacement (Benally et al., 2022), as shown in Merry's gesture towards Philip (Figure 3C) and Philip's line tracing from Merry to his spot (Figure 3D). Owen, whose strategy had been unknown, also affirmed *opposites* by tracing a line to Nath (Figure 3E).

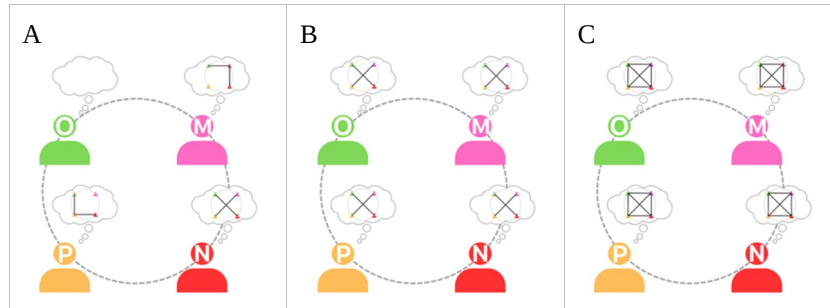
Figure 4: Students inscribe their auxiliary reference lines, with overhead insets



Resolved. When prompted to draw their reference lines, Nath, Owen, and Philip began with *opposites* (Figure 4A–B), while Merry reasserted *adjacents* by connecting a line to Nath (Figure 4C). Seeing Merry's adjacent line, Owen drew the respective line from his location to Merry (Figure 4D) and then to Philip (Figure 4E). In the final panel, all students drew the line connecting Philip and Nath (Figure 4F), bringing both strategies

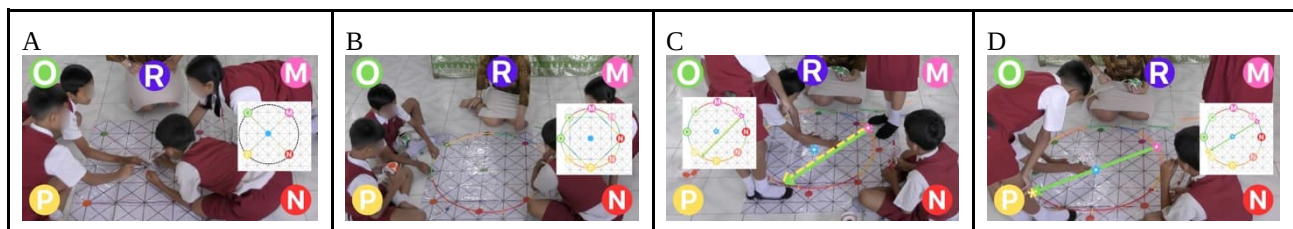
into a shared diagram. Through this integration of previously distinct strategies, the group achieved perspectival synergy (Benally et al., 2022). Figure 5 summarizes the progression of heteroperception from implicit to explicit to resolved.

Figure 5: Progression of heteroperception stages in the GRiD make-a-circle study



Prompted by the researcher, the group drew on their attentional anchors to further investigate the elements of the circle. First, they identified the center point of the constructed circle by indicating the intersection of their *opposites* attentional anchors (Figure 6A). This exploration led students to gauge that they were all equidistant from the center (radius) and that the distance between opposite pairs was twice that of an individual to the center (diameter). Once the group built their circle using straws and nodes (Figure 6B)—thus materially realizing their collective imaginary (Palatnik & Abrahamson, in press)—their discussion extended to arbitrary locations along the circumference, specifically addressing where each person should stand in the formation when one participant moves. Informed by the dance context (recall Figure 1D–E), when Merry was asked to relocate, other students adjusted their positions accordingly, maintaining the *adjacents* AA (Figure 6B).

Figure 6: Students inspect the elements of a circle, with overhead insets



Note. Panel A: Students mark the center of the circle formation they formed. Panel B–D: Students construct the circle using straws and nodes, then explore new locations, as one participant (Merry) moves along the circumference. Initial positions are shown in faded color, second positions in solid color, and third positions are indicated by flower icons.

Next, students were challenged to explore a counterintuitive case, in which a pre-existing line on the GRiD mat no longer supported their reasoning. Merry was once again repositioned, this time facing a chord that is not a diameter (Figure 6C, pink flower). Initially unaware of the new circumstance, Philip and Owen followed the available line on the mat (Figure 6C). Merry immediately challenged their construction, emphasizing that “*it should pass through this point, the center point!*” In response, Owen revised his strategy by drawing a new auxiliary line from Merry’s updated position, through the center point, and extending it to the opposite side of the

circle to determine Philip's new location (Figure 6D). The group then continued exploring across multiple configurations, each time with a different student initiating the movement, which enabled the group to conclude that a circle consists of infinitely many points, each of which admits a corresponding opposite point through the center.

CONCLUSION AND IMPLICATION

Students set out with differing yet complementary perceptions (implicit). Through discussion, some recognized this heteroperception and substituted their strategy (explicit). When later externalized in shared space, the initial strategy was reintroduced, resulting in synergy between different perceptions (resolved). Importantly, heteroperception in this study was not hierarchical; rather, alternative perceptions equitably supported the group's collective effort to enact and define a circle. We therefore propose that heteroperception in collaborative mathematical problem-solving can function as a generative resource for learning by stimulating the social reification of tacit perceptual orientations. Heteroperception should be anticipated, intentionally elicited, valorized, and supported through instructional facilitation. Finally, visualizing ensemble learning processes through the lens of heteroperception helps analysts appreciate, monitor, and theorize the emergence of mathematical discourse from situated movement problems.

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